

6.15 STATISTICS 3, S3 (4768) A2

Objectives

To introduce students to more advanced statistical ideas, based on the foundation provided by *S1* and *S2*, particularly involving continuous random variables and inference.

Assessment

Examination (72 marks)
 1 hour 30 minutes
 There will be four questions each worth about 18 marks.

Assumed Knowledge

Candidates are expected to know the content for *C1*, *C2*, *C3* and *C4* and for *S1* and *S2*. In addition they may be required to evaluate integrals of the type $\int_a^{\infty} f(x)dx$.

Calculators

In the MEI Structured Mathematics specification, no calculator is allowed in the examination for *C1*. For all other units, including this one, a graphical calculator is allowed.

STATISTICS 3, S3		
Specification	Ref.	Competence Statements

CONTINUOUS RANDOM VARIABLES

The probability density function (pdf) of a continuous random variable.	S3R1	Be able to use a simple continuous random variable as a model.
	2	Understand the meaning of a pdf and be able to use one to find probabilities.
	3	Know the properties of a pdf.
	4	Find the mean and variance from a given pdf.
	5	Find the mode and median from a given pdf.
The cumulative distribution function (cdf) and its relationship to the probability density function.	6	Understand the meaning of a cdf and know how to obtain one from a given pdf.
	7	Know how to obtain a pdf from a given cdf.
	8	Use a cdf to calculate the median and other percentiles.

EXPECTATION ALGEBRA

Calculation of probability, expectation (mean) and variance of random variables.	S3a1	Be able to use the result $E(a + bX) = a + bE(X)$.
	2	Be able to use the result $\text{Var}(a + bX) = b^2\text{Var}(X)$.
Linear combinations of two (or more) independent random variables.	3	Know how to find the mean and variance of any linear combination of independent random variables.
The distribution of a linear combination of independent Normal variables.	4	Be able to use linear combinations of Normal random variables in solving problems.

STATISTICS 3, S3			
Ref.	Notes	Notation	Exclusions

CONTINUOUS RANDOM VARIABLES

S3R1			
2	Concepts rather than expertise at calculus will be examined.	$f(x)$	pdfs comprising more than two (non-zero) parts.
3			
4			
5			
6		$F(x)$	
7	$f(x) = F'(x)$		
8			

EXPECTATION ALGEBRA

S3a1			
2			
3	$E(X \pm Y) = E(X) \pm E(Y)$ $\text{Var}(X \pm Y) = \text{Var}(X) + \text{Var}(Y)$ $E(aX \pm bY) = a E(X) \pm b E(Y)$ $\text{Var}(aX \pm bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y)$		Proofs.
4	A linear combination of independent normal variables so also normal.		

STATISTICS 3, S3		
Specification	Ref.	Competence Statements

INFERENCE

Sampling methods.	S3I1	Know the definition of the term simple random sample.
	2	Understand the use of different sampling methods.
Estimation of population mean and variance from a simple random sample.	3	Be able to estimate population mean from sample data.
	4	Be able to estimate population variance using the sample variance, s^2 .
Distribution of the mean of a sufficiently large sample.	5	Understand how and when the Central Limit Theorem may be applied to the distribution of sample means.
Standard error of the mean.	6	Be able to calculate and interpret the standard error of the mean.
Symmetric confidence intervals for the mean.	7	Understand the term confidence interval and be able to construct confidence intervals for the mean using Normal or t distributions.
	8	Be able to construct and interpret confidence intervals for the difference in mean of paired populations using the Normal or t distribution.
Hypothesis tests: (a) t test for a single mean;	9	Be able to carry out a hypothesis test for a single mean using the t distribution and know when it is appropriate to do so.
(b) paired t test;	10	Be able to carry out a hypothesis test on the difference of means for paired data using the t distribution and know when it is appropriate to do so.
(c) Wilcoxon signed rank test;	11	Be able to carry out a hypothesis test for a single median using the Wilcoxon signed rank test and know when it is appropriate to do so.
(d) Wilcoxon signed rank test;	12	Be able to carry out a hypothesis test for paired data using the Wilcoxon signed rank test and know when it is appropriate to do so.
(e) χ^2 test for goodness of fit.	13	Be able to carry out a χ^2 test for goodness of fit of a model, understanding and using degrees of freedom.

STATISTICS 3, S3			
Ref.	Notes	Notation	Exclusions
INFERENCE			
S311	A simple random sample is one in which every possible sample of the given size has an equal probability of being selected.		
2	Simple random sampling, systematic sampling, stratified sampling, cluster sampling, quota sampling (including opportunity sampling).		
3		$\hat{\mu} = \bar{x}$	Proof.
4	$s^2 = \frac{1}{(n-1)} \sum_{i=1}^n (x_i - \bar{x})^2$		Proof.
5	The implication of the Central Limit Theorem.		Formal statement and derivation of the Central Limit Theorem.
6	The standard error of the mean is $\frac{\sigma}{\sqrt{n}}$, if σ is known; otherwise it is estimated as $\frac{s}{\sqrt{n}}$.		
7	In situations where: (a) the population variance is known (Normal); (b) the population variance is unknown but the sample size is large (Normal); (c) the sample is small and the population variance is unknown, but the population may be assumed to have a Normal distribution (t).		
8			
9	In situations where the sample is small and the population variance is unknown, but the population may be assumed to have a Normal distribution.		
10			
11			
12			
13	e.g. uniform distribution, binomial distribution, Poisson distribution, Normal distribution.		